

THE FIVE-COLOR CASE OF AN ERDŐS–GYÁRFÁS BALANCED-COLORING PROBLEM

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ABSTRACT. We prove the five-color case of a conjecture of Erdős and Gyárfás: every five-coloring of the edges of K_{26} has six vertices whose induced edges omit at least one color. Equivalently, the set-coloring Ramsey number $R(6; 5, 4)$ equals 26. The proof assigns a graph to each color, applies the Kang–Pikhurko extremal theorem to complements of induced subgraphs, and combines the resulting edge bounds with minimum-degree decompositions. A sequence of small near-extremal lemmas first forces all five color graphs to have 65 edges and then excludes the equality case. The argument is non-computational; finite searches used during its development are not premises of the proof.

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1. INTRODUCTION

Erdős and Gyárfás asked whether the following assertion holds for every integer $r \geq 3$: whenever the edges of K_{r^2+1} are colored with r colors, some $r + 1$ vertices induce a complete graph on which at least one color is absent. They proved the cases $r = 3$ and $r = 4$, observed that the analogous assertion fails for $r = 2$, and studied the related problem on r^2 vertices [5]. The problem also appears in Erdős’s later problem collection [4] and as Erdős Problem 617 [1].

The purpose of this paper is to prove the assertion for $r = 5$.

Theorem 1.1. *For every map $\chi : E(K_{26}) \rightarrow [5]$, there is a set $S \subseteq V(K_{26})$ with $|S| = 6$ such that*

$$\chi(E(K_{26}[S])) \neq [5].$$

This is a fixed-parameter result. In particular, Theorem 1.1 does not settle the conjecture for all $r \geq 3$.

There is an equivalent formulation in the language of set-coloring Ramsey numbers. An (r, s) -coloring assigns to each edge an s -element subset of a

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palette of r colors. The number $R(n; r, s)$ is the least N for which every (r, s) -coloring of K_N has a copy of K_n whose edges share a common color [3]. Replacing the color on each edge by the other four colors turns Theorem 1.1 into the upper bound $R(6; 5, 4) \leq 26$.

The reverse bound has a short affine construction. Take the vertex set $\mathbb{F}_5^2$. Its six parallel classes of affine lines are indexed by the slopes $\mathbb{F}_5 \cup \{\infty\}$. Color a pair of points by the slope of the line through them, after merging the classes of slopes 0 and ∞ into one color. For each of the four unmerged slopes, any six points contain two on a common line of that slope, by the pigeonhole principle applied to its five parallel lines. The same argument with the horizontal lines supplies the merged color. Thus every six points see all five colors. Complementing the edge colors gives a $(5, 4)$ -coloring of K_{25} with no monochromatic K_6 .

Corollary 1.2. $R(6; 5, 4) = 26$.

We now describe the proof of Theorem 1.1. Assuming a counterexample, let G_i be the graph formed by color i . Every six-set spans between one and eleven edges in each G_i . The Kang–Pikhurko theorem first gives a uniform lower bound on $e(G_i)$ and confines a least color to minimum degree two, three, or four. We then prove three layers of induced-subgraph obstructions, for independence numbers two, three, and four. The last layer forces $e(G_i) \geq 65$ for every i , so all five color graphs have exactly 65 edges. Equality in the minimum-degree counts repeatedly peels off isolated cliques. The remaining near-extremal graphs are excluded by two explicit classifications on fifteen vertices.

2. PRELIMINARIES

All graphs are finite and simple. We write $v(F) = |V(F)|$, $e(F) = |E(F)|$, $\alpha(F)$ for the independence number, $\omega(F)$ for the clique number, $\tau(F)$ for the minimum vertex-cover number, and $\delta(F)$ and $\Delta(F)$ for the minimum and maximum degrees. The complement of F is $\overline{F}$, and $F[W]$ is the subgraph induced by W . The Turán graph with r parts on n vertices is $T_r(n)$, and $t_r(n) = e(T_r(n))$.

Suppose, for the rest of the proof, that a counterexample to Theorem 1.1 is given. For a fixed color, let G be its color graph. For every $S \in \binom{V(G)}{6}$,

$$1 \leq e(G[S]) \leq 11. \quad (1)$$

Indeed, the lower bound says that the chosen color is present on S , while the upper bound follows because each of the other four colors occupies at least one of the fifteen edges. In particular,

$$\alpha(G) \leq 5, \quad \omega(G) \leq 5. \quad (2)$$

Every induced subgraph of a color graph retains the upper bound in (1).

Definition 2.1. A graph is *admissible* if every six vertices span at most eleven edges.

The main external extremal result is the following theorem of Kang and Pikhurko [6, Theorems 1 and 4, Lemma 5].

Theorem 2.2 (Kang–Pikhurko). *Let $n \geq r + 3$, $r \geq 2$, and let L be a K_{r+1} -free graph on n vertices that is not r -partite. If $r \leq (n - 1)/2$, then*

$$e(L) \leq t_r(n) - \left\lfloor \frac{n}{r} \right\rfloor + 1. \quad (3)$$

Equality holds only for the graphs in the construction described below.

For the equality construction, choose positive integers $n_1 \leq \dots \leq n_r$ with sum $n - 1$ and at least two entries larger than one. Let $N_1, \dots, N_r$ be disjoint sets of these sizes, and let s, t , in either order, be the two smallest indices whose parts have size greater than one. Choose a proper nonempty set $A \subset N_s$ and a vertex $y \in N_t$. Begin with the complete r -partite graph on the N_i , add a vertex x adjacent to

$$\bigcup_{i \notin \{s, t\}} N_i \cup A \cup \{y\},$$

and delete all edges between y and A . Kang and Pikhurko prove that all equality graphs arise this way, with a short list of optimal part sizes. The instances needed below are

r	n	optimal sequences $(n_1, \dots, n_r)$
2	10	(4, 5)
3	15	(4, 4, 6), (4, 5, 5)

We will repeatedly exclude equality in (3) using (1). The required checks are recorded once here.

Lemma 2.3 (Local exclusion of equality). *Let L be a K_{r+1} -free graph on n vertices that is not r -partite, and let $R = \overline{L}$ be admissible.*

- (i) *If $(r, n) = (2, 10)$, then $e(L) \leq 20$.*
- (ii) *If $(r, n) = (3, 15)$, equality in (3) is impossible.*

Proof. For (i), the unique sequence is (4, 5). If the part of size five is the t -part, that part together with x spans fourteen edges in R . If it is the s -part and $|A| \leq 3$, the same six-set spans at least twelve edges. If $|A| = 4$, use the size-five part together with y ; it spans fourteen edges.

For (ii), the sequence (4, 4, 6) gives a K_6 in R . Consider (4, 5, 5). If a size-five t -part is used, that part with x spans fourteen edges in R . If a size-five s -part is used and $|A| \leq 3$, that part with x spans at least twelve edges. If $|A| = 4$, the size-five s -part with y spans fourteen edges. $\square$

We also use Brooks's theorem in the standard form: a connected graph of maximum degree D is D -colorable unless it is complete or an odd cycle [2].

Lemma 2.4 (Neighborhood accounting). *Let F be a graph, let v be a minimum-degree vertex of degree d , let $N = N_F(v)$, and let $U = V(F) \setminus (N \cup \{v\})$. Put*

$$X = e_F(N, U), \quad Y = e(F[N]), \quad A = X + Y.$$

Then

$$X + 2Y \geq d(d - 1), \quad Y \leq \binom{d}{2}, \quad A \geq \binom{d}{2}. \quad (4)$$

If F is admissible and $d = 5$, then $Y \leq 6$ and $A \geq 14$. If equality $A = \binom{d}{2}$ holds in the last inequality of (4), then $N \cup \{v\}$ is an isolated K_{d+1} in F .

Proof. Every vertex of N has at least $d - 1$ incident edges other than its edge to v . Summing these requirements over N counts the N - U edges once and the edges inside N twice, proving the first inequality. Combining it with $Y \leq \binom{d}{2}$ gives the third. When $d = 5$, the six-set $N \cup \{v\}$ has $5 + Y \leq 11$ edges. Finally, $X + Y = \binom{d}{2}$ and $X + 2Y \geq d(d - 1) = 2\binom{d}{2}$ force $Y = \binom{d}{2}$ and $X = 0$. $\square$

Proposition 2.5 (Initial structural reduction). *If G is a least frequent color graph, then*

$$e(G) \leq 65, \quad 2 \leq \delta(G) \leq 4.$$

Proof. The five edge counts sum to $\binom{26}{2} = 325$, so a least color has at most 65 edges. If its minimum degree were at least five, its average degree would force it to be five-regular. Each component is neither K_6 nor an odd cycle: K_6 is excluded by (2), and an odd cycle is not five-regular. Brooks's theorem gives a proper five-coloring. One color class would have at least six vertices, contrary to $\alpha(G) \leq 5$. Hence $\delta(G) \leq 4$.

Let v be a minimum-degree vertex of degree d . Its $25 - d$ nonneighbors induce a graph $G[U]$ with independence number at most four. Thus $\overline{G[U]}$ is K_5 -free. When $d = 0$ or 1 , this complement is not four-partite: a four-partition of 25 or 24 vertices has a part of size at least six, which would be a forbidden clique in $G[U]$. For $d = 0$, Theorem 2.2 gives $e(G[U]) \geq 71$. For $d = 1$, the same theorem on 24 vertices gives $e(G[U]) \geq 65$, and the edge incident with v gives $e(G) \geq 66$. Both cases contradict $e(G) \leq 65$, so $d \geq 2$. $\square$

3. TERMINAL BOUNDS AT INDEPENDENCE NUMBER TWO

The first layer concerns complements of triangle-free graphs. Notice that if an admissible graph F has $\alpha(F) \leq 2$, then $L = \overline{F}$ is triangle-free and every six vertices span at least four edges in L .

Lemma 3.1. *Let F be admissible with $\alpha(F) \leq 2$ and $v(F) = n \geq 12$. Then*

$$e(F) \geq \binom{n}{2} - 2n.$$

Proof. Put $L = \overline{F}$. A vertex of degree at least six in L has six independent neighbors, a contradiction. Suppose that w has degree five. Let $A = N_L(w)$ and let B be the remaining $n - 6$ vertices. The set A is independent. For each $x \in B$, the six-set $A \cup \{x\}$ forces $d_A(x) \geq 4$. On the other hand, each vertex of A uses one incident edge on w and has total degree at most five, so

$$e_L(A, B) \leq 5 \cdot 4 = 20.$$

Since $n \geq 12$, the lower bound $4|B| \geq 24$ is impossible. Therefore $\Delta(L) \leq 4$ and $e(L) \leq 2n$. $\square$

The order eleven endpoint needs a sharper argument.

Lemma 3.2. *Let F be an admissible graph on eleven vertices with $\alpha(F) \leq 2$. Then $e(F) \geq 36$.*

Proof. Put $L = \overline{F}$. As above, L is triangle-free, every six-set spans at least four L -edges, and $\Delta(L) \leq 5$. A degree-five vertex is also impossible. Indeed, if $A = N_L(w)$ and B is the other five vertices, each $x \in B$ has at least four neighbors in A . Two vertices of B cannot be adjacent, because their two subsets of at least four neighbors in the five-set A intersect and would form a triangle. Thus $B \cup \{w\}$ is an independent six-set in L , contrary to the four-edge condition. Hence $\Delta(L) \leq 4$.

Suppose that $e(F) \leq 35$, so $e(L) \geq 20$. Choose a degree-four vertex w , put $A = N_L(w)$, and let B be the remaining six vertices. For $b \in B$, write $p_b = d_A(b)$. For distinct $x, y \in B$, the six-set $A \cup \{x, y\}$ gives

$$p_x + p_y + \mathbf{1}_{xy \in E(L)} \geq 4. \quad (5)$$

Also,

$$\sum_{b \in B} p_b = e_L(A, B) \leq 4 \cdot 3 = 12. \quad (6)$$

These inequalities force $p_b = 2$ for every $b \in B$. To verify this, order the six values. Two values at most one violate (5); a zero forces the other five values to be at least three, contradicting (6). If one value is one, all others are at least two. The only remaining totals at most twelve are $(1, 2, 2, 2, 2, 2)$ and $(1, 2, 2, 2, 2, 3)$. The vertex with value one must then be adjacent in B to at least four vertices with value two, although its B -degree is at most three.

It follows that every vertex of A has degree four and every vertex of B has B -degree at most two. Also

$$e(L[B]) = e(L) - 4 - 12.$$

If $e(L) \geq 23$, this exceeds the maximum possible number of edges in $L[B]$. For $e(L) = 20, 21, 22$, the graph $L[B]$ has respectively four, five, or six edges. Along an edge of $L[B]$, the two 2-subsets of A given by the endpoint neighborhoods are disjoint, hence complementary.

If $L[B]$ has six edges, triangle-freeness and maximum degree two force $L[B] = C_6$. Its two parity classes carry alternating complementary labels P and $A \setminus P$. The set consisting of w , the two vertices of A belonging to P , and the three vertices in the parity class labeled $A \setminus P$ spans only the two edges from w to P .

If $L[B]$ has five edges, its triangle-free possibilities are P_6 , C_5 plus an isolated vertex, and C_4 plus P_2 . The odd cycle is incompatible with alternating complementary labels. The path gives the same parity-class contradiction as C_6 . In the last case, take two independent cycle vertices with one label. Of the endpoints of P_2 , choose one whose label meets the complementary 2-set in at most one point. Together with w and that complementary 2-set, these vertices span at most three edges.

If $L[B]$ has four edges, its triangle-free possibilities are P_5 plus an isolated vertex, P_4 plus P_2 , two copies of P_3 , and C_4 plus two isolated vertices. The path case is again the parity argument. For the next two cases, take two independent same-label vertices in one path and a vertex from the other component whose label meets the complementary 2-set in at most one point. In the C_4 case, use two opposite same-label cycle vertices and a suitable isolate. If neither isolate is suitable, both carry the complementary label,

and the two isolates with a cycle vertex of that label give the required set. In every case there is a six-set with at most three L -edges, a contradiction. $\square$

We will also need the structure at order ten.

Lemma 3.3 (Ten-vertex structure). *Let L be triangle-free on ten vertices, and suppose every six vertices span at least four edges.*

- (i) *If L is bipartite, its parts both have size five. In $\bar{L}$ the two parts are K_5 's and the cross-edges form a matching.*
- (ii) *If L is nonbipartite, then $e(L) \leq 20$.*
- (iii) *If L is nonbipartite and $e(L) = 20$, then L is the balanced two-fold blow-up of C_5 .*
- (iv) *If $e(L) = 19$, then $\alpha(L) \leq 4$.*

For the balanced two-fold blow-up T , one has $\alpha(T) = 4$ and $\tau(T) = 6$. Its minimum vertex covers are unions of three of the five parts, and two distinct minimum covers intersect in at most four vertices.

Proof. If L is bipartite, each part has size at most five, since an independent six-set is forbidden. Thus both parts have size five. A vertex of one K_5 in $\bar{L}$ has at most one neighbor in the other, since that vertex together with the first K_5 would otherwise span at least twelve complement edges. This proves (i).

For nonbipartite L , Theorem 2.2 gives $e(L) \leq 21$, and Lemma 2.3(i) excludes equality. This proves (ii).

Suppose $e(L) = 20$. A degree-five vertex would have an independent five-neighborhood A . Each of the other four vertices would need at least four neighbors in A , and triangle-freeness would make those four vertices independent. This would require at least $5 + 16 = 21$ edges. Hence L is four-regular. Take a shortest odd cycle C of length ℓ . It is induced, and the 2ℓ edges from C to its complement fit into only $4(10 - \ell)$ outside degree slots, so $\ell \leq 6$. Triangle-freeness and oddness give $\ell \geq 5$ odd, and hence $\ell = 5$. Each outside vertex has exactly two nonconsecutive neighbors on C . If x_i counts the outside vertices adjacent to cycle vertices $i - 1$ and $i + 1$, the external degree two of cycle vertex j gives

$$x_{j-1} + x_{j+1} = 2.$$

The recurrence around the odd cycle forces $x_i = 1$ for every i . The outside graph is two-regular and triangle-free, hence another C_5 , and its allowed edges join consecutive types. Pair cycle vertex i with the unique outside vertex of type i . Each pair is independent, and every two consecutive pairs are joined completely. Four-regularity leaves no other edges, so these five pairs exhibit L as the balanced two-fold blow-up of C_5 . This proves (iii).

For (iv), an independent five-set would force each of the remaining five vertices to have at least four neighbors in it, already accounting for twenty edges. Finally, in the balanced blow-up the maximum independent sets are precisely the unions of two nonconsecutive size-two parts. The claims about covers follow by taking complements. $\square$

4. OBSTRUCTIONS AT INDEPENDENCE NUMBER THREE

The terminal estimates propagate to the next independence level.

Lemma 4.1. *There is no admissible graph F with $\alpha(F) \leq 3$ in any of the following ranges:*

$$(v(F), e(F)) \in \{(16, \leq 44), (17, \leq 49), (18, \leq 53)\}.$$

Proof. First suppose that $v(F) = 16$ and $e(F) \leq 44$. A minimum-degree vertex v has degree $d \leq 5$. For $d \leq 4$, the graph on its $15 - d$ nonneighbors has independence number at most two. Lemmas 3.1 and 3.2, together with Lemma 2.4, give the lower bounds

d	0	1	2	3	4
$e(F)$	75	$63 + 1$	$52 + 2 + 1$	$42 + 3 + 3$	$36 + 4 + 6$

and every entry exceeds 44.

It remains to consider $d = 5$. Use the notation of Lemma 2.4, put $R = F[U]$, and put $L = \overline{R}$. Then

$$e(R) \leq 44 - 5 - 14 = 25, \quad e(L) \geq 20. \quad (7)$$

If L is bipartite, Lemma 3.3(i) says that R consists of two K_5 's with at most five cross-edges, which form a matching. A vertex of N has at most one F -neighbor in either clique. If two vertices of N were nonadjacent, each clique would contain three vertices avoiding both, and some pair across the two three-sets would be a nonedge of R . This would give an independent four-set in F . Thus $N = K_5$, and $N \cup \{v\}$ is a forbidden K_6 .

Suppose instead that L is nonbipartite. Lemma 3.3(ii) and (7) force equality throughout:

$$e(F) = 44; \quad e(R) = 25; \quad e(L) = 20; \quad A = 14; \quad Y = 6; \quad X = 8.$$

By Lemma 3.3(iii), L is the balanced two-fold blow-up T of C_5 . Let $M = \overline{F[N]}$. Then $e(M) = 4$. If x_a denotes the number of F -neighbors of $a \in N$ in U , minimum degree gives $x_a \geq d_M(a)$. Since both sides sum to eight, equality holds for every a .

For an edge $aa' \in E(M)$, the two U -neighborhoods must cover every edge of T ; otherwise an uncovered edge of T , together with a, a' , is an independent four-set in F . On the other hand,

$$x_a + x_{a'} = d_M(a) + d_M(a') \leq e(M) + 1 = 5,$$

because $d_M(a) + d_M(a')$ counts every edge incident with a or a' once and counts aa' a second time, whereas $\tau(T) = 6$ by Lemma 3.3. This contradiction settles the sixteen-vertex case.

For the remaining orders the same minimum-degree decomposition is purely numerical. The complete tables are

	$d = 0$	$d = 1$	$d = 2$	$d = 3$	$d = 4$	$d = 5$
$v(F) = 17$	88	$75 + 1$	$63 + 2 + 1$	$52 + 3 + 3$	$42 + 4 + 6$	$36 + 5 + 14$
$v(F) = 18$	102	$88 + 1$	$75 + 2 + 1$	$63 + 3 + 3$	$52 + 4 + 6$	$42 + 5 + 14$

Every entry in the first row exceeds 49, and every entry in the second exceeds 53. $\square$

We also record four direct consequences of Theorem 2.2.

Lemma 4.2. *If F is admissible, $\alpha(F) \leq 3$, and $19 \leq v(F) \leq 22$, then*

$v(F)$	19	20	21	22
$e(F)$	≥ 56	≥ 62	≥ 69	≥ 76

Proof. Put $L = \overline{F}$. It is K_4 -free. It is not three-partite, since a part would have at least six vertices and would give a K_6 in F . Subtracting the bound (3) from $\binom{v(F)}{2}$ gives the four displayed values. $\square$

5. NEAR-EXTREMAL GRAPHS ON FIFTEEN VERTICES

Two small structure statements will be used in the last step. In both, the assumption that $\overline{R}$ is not three-partite is equivalent to saying that $V(R)$ cannot be covered by three cliques.

Lemma 5.1. *Let R be an admissible graph on fifteen vertices with $\alpha(R) \leq 3$, and suppose that $\overline{R}$ is not three-partite. If $e(R) = 35$, then*

$$R = K_5 \dot{\cup} R_0,$$

where R_0 has ten vertices and $\overline{R_0}$ is the balanced two-fold blow-up of C_5 .

Proof. The average degree is less than five. A minimum-degree vertex cannot have degree at most three: the graph on its nonneighbors has independence number at most two, and for degrees zero, one, two, and three the bounds from Section 3, including the vertex and neighborhood contributions, are respectively

$$63, \quad 52 + 1, \quad 42 + 2 + 1, \quad 36 + 3 + 3,$$

all larger than 35. Hence $\delta(R) = 4$.

Choose a degree-four vertex w , let $N = N_R(w)$, and let Q be its ten nonneighbors. Put $X = e_R(N, Q)$, $Y = e(R[N])$, and $B = X + Y$. Then

$$e(R[Q]) + B = 31, \quad B \geq 6, \quad e(R[Q]) \leq 25. \quad (8)$$

Also $\alpha(R[Q]) \leq 2$, since an independent triple in Q together with w would be an independent four-set. Thus $T = \overline{R[Q]}$ is triangle-free and every six vertices span at least four T -edges.

Suppose $e(R[Q]) \leq 24$, so $e(T) \geq 21$. If T is nonbipartite, Theorem 2.2 and Lemma 2.3(i) give $e(T) \leq 20$, a contradiction. Hence T is bipartite. Its parts both have size five, and $R[Q]$ consists of two K_5 's with at most four matching cross-edges. Each vertex of N has at most one neighbor in either clique. If two vertices of N were nonadjacent, three vertices in each clique would avoid both, and some cross-pair among them would be a nonedge of R . This gives an independent four-set. Therefore $N = K_4$, and R is covered by the three cliques $N \cup \{w\}$ and the two cliques in Q , contrary to the hypothesis.

It follows from (8) that $e(R[Q]) = 25$ and $B = 6$. Lemma 2.4 now gives $Y = 6$ and $X = 0$, so $N \cup \{w\}$ is an isolated K_5 . The graph T is not bipartite, since otherwise the two cliques in Q and this K_5 would cover R . It has twenty edges, and Lemma 3.3(iii) identifies it as the balanced two-fold blow-up of C_5 . $\square$

Lemma 5.2. *Let R be an admissible graph on fifteen vertices with $\alpha(R) \leq 3$, and suppose that $\overline{R}$ is not three-partite. If $e(R) = 36$, then exactly one of the following holds.*

- (i) *There is a partition $V(R) = C \dot{\cup} V(R_0)$ with $|C| = 5$, $R[C] = K_5$, and exactly one edge between C and $V(R_0)$, where $\overline{R_0}$ is the balanced two-fold blow-up of C_5 .*
- (ii) *The graph R is an isolated K_5 together with a ten-vertex graph R_0 such that $\overline{R_0}$ is triangle-free, has nineteen edges, and has independence number at most four.*

Proof. As in the proof of Lemma 5.1, the minimum degree is four. With the same notation,

$$e(R[Q]) + B = 32, \quad B \geq 6, \quad e(R[Q]) \leq 26. \quad (9)$$

The values $e(R[Q]) \leq 24$ give the same three-clique-cover contradiction.

Suppose first that $e(R[Q]) = 25$. The triangle-free graph $T = \overline{R[Q]}$ has twenty edges. If it were bipartite, the same candidate argument would force $N = K_4$ and again give a three-clique cover. Hence T is the balanced two-fold blow-up of C_5 . Here $B = 7$, and

$$B + Y = X + 2Y \geq 12,$$

so $Y \geq 5$. If $Y = 5$, the graph $R[N]$ is K_4 minus one edge. The two nonadjacent endpoints have at most $X = 2$ neighbors in Q , but their Q -neighborhoods must cover every edge of T to avoid an independent four-set. This contradicts $\tau(T) = 6$. Therefore $Y = 6$ and $X = 1$, which is alternative (i).

Finally suppose that $e(R[Q]) = 26$. Then $B = 6$, and Lemma 2.4 gives $Y = 6$ and $X = 0$. Thus $N \cup \{w\}$ is an isolated K_5 . The graph T has nineteen edges. It cannot be bipartite: two parts of size five would make $R[Q]$ two K_5 's with six cross-edges, although those cross-edges form a matching and hence number at most five. Lemma 3.3(iv) gives $\alpha(T) \leq 4$. This is alternative (ii). $\square$

6. OBSTRUCTIONS AT INDEPENDENCE NUMBER FOUR

Lemma 6.1. *There is no admissible graph F with $\alpha(F) \leq 4$ in any of the following ranges:*

$$(v(F), e(F)) \in \{(23, \leq 61), (22, \leq 58), (21, \leq 54)\}.$$

Proof. In each range the average degree is less than six, so a minimum-degree vertex has degree at most five. Its nonneighbors induce a graph with independence number at most three. Lemmas 4.1 and 4.2, together with Lemma 2.4, give the following complete lower-bound table for the first two orders:

	$d = 0$	$d = 1$	$d = 2$	$d = 3$	$d = 4$	$d = 5$
$v(F) = 23$	76	$69 + 1$	$62 + 2 + 1$	$56 + 3 + 3$	$54 + 4 + 6$	$50 + 5 + 14$
$v(F) = 22$	69	$62 + 1$	$56 + 2 + 1$	$54 + 3 + 3$	$50 + 4 + 6$	$45 + 5 + 14$

Every entry in the first row exceeds 61, and every entry in the second exceeds 58.

For $v(F) = 21$, degrees zero through four are also excluded numerically. The closest three bounds are

$$54 + 2 + 1 = 57, \quad 50 + 3 + 3 = 56, \quad 45 + 4 + 6 = 55$$

for degrees two, three, and four respectively. It remains to consider $d = 5$. Let v be such a vertex, let $N = N_F(v)$, let U be its fifteen nonneighbors, put $R = F[U]$, and put $L = \overline{R}$. Then

$$e(R) \leq 54 - 5 - 14 = 35. \quad (10)$$

Suppose first that L is three-partite. Each part has size five: a part is a clique in R , and admissibility forbids a clique of size six. Thus R consists of three K_5 's and at most five cross-edges. Every vertex of N has at most one neighbor in each clique. If $a, a' \in N$ were nonadjacent, each clique would supply at least three vertices avoiding both. There are $3^3 = 27$ triples choosing one such vertex from each clique, while each cross-edge rules out at most three triples. Since $5 \cdot 3 < 27$, an independent triple in R remains. Together with a, a' it is an independent five-set in F . Therefore $N = K_5$, and $N \cup \{v\}$ is a forbidden K_6 .

Hence L is not three-partite. By Theorem 2.2, we have $e(R) \geq 34$. Lemma 2.3(ii) excludes equality. Thus $e(R) = 35$, and equality holds in all remaining counts:

$$e(F) = 54; \quad A = 14; \quad Y = 6; \quad X = 8.$$

Let $M = \overline{F[N]}$. Then $e(M) = 4$, and, as in the proof of Lemma 4.1, each $a \in N$ has exactly $d_M(a)$ neighbors in U .

By Lemma 5.1,

$$R = C \cup R_0,$$

where $C = K_5$ and $T = \overline{R_0}$ is the balanced two-fold blow-up of C_5 . Choose an edge $aa' \in E(M)$. Each endpoint has at most one neighbor in C , so some $c \in C$ avoids both. If an edge of T avoids the R_0 -neighborhoods of a, a' , that edge together with a, a', c is an independent five-set in F . Thus those two neighborhoods cover T , but their total size is at most

$$d_M(a) + d_M(a') \leq e(M) + 1 = 5 < \tau(T),$$

a contradiction. □

7. EDGE EQUALIZATION AND THE FINAL REDUCTION

The preceding obstruction removes the need to analyze separate cases with sixty, sixty-one, sixty-two, sixty-three, and sixty-four edges.

Proposition 7.1. *Every color graph has exactly 65 edges.*

Proof. Let G be a least frequent color graph. By Proposition 2.5, $e(G) \leq 65$ and a minimum-degree vertex has degree $d \in \{2, 3, 4\}$. With the notation of Lemma 2.4,

$$e(G[U]) = e(G) - d - A \leq e(G) - d - \binom{d}{2}. \quad (11)$$

If $e(G) \leq 64$, the right side is at most 61, 58, or 54 for $d = 2, 3$, or 4. The graph $G[U]$ has respectively 23, 22, or 21 vertices, is admissible, and has independence number at most four. Lemma 6.1 excludes all three possibilities. Hence $e(G) = 65$. Since the five edge counts sum to $\binom{26}{2} = 325 = 5 \cdot 65$, every color graph has 65 edges. $\square$

Every color is now tied for least frequent, so Proposition 2.5 applies to each color graph.

Lemma 7.2. *No color graph has minimum degree two or three.*

Proof. Let G be a color graph. Suppose first that G has a degree-two vertex. Equation (11) and Lemma 6.1 force

$$e(G[U]) = 62, \quad A = 1, \quad Y = 1, \quad X = 0.$$

Thus $G = K_3 \dot{\cup} H$, where H has 23 vertices, 62 edges, minimum degree at least two, and independence number at most four.

The average degree of H is less than six. At a minimum-degree vertex of degree two, four, or five, the graph on its nonneighbors has independence number at most three, and Lemmas 4.1 and 4.2 give respective total lower bounds

$$62 + 2 + 1, \quad 54 + 4 + 6, \quad 50 + 5 + 14,$$

all larger than 62. Therefore $\delta(H) = 3$. The nineteen nonneighbors then induce at least 56 edges, so equality is forced:

$$e(H[U]) = 56, \quad A = 3, \quad Y = 3, \quad X = 0.$$

Peeling off the resulting isolated K_4 leaves a graph Q on nineteen vertices with 56 edges, minimum degree at least three, and independence number at most three. Its average degree is less than six. At a minimum-degree vertex of degree three, four, or five, the nonneighbor graph has independence number at most two. Lemma 3.1 and the neighborhood contributions give the respective totals

$$75 + 3 + 3, \quad 63 + 4 + 6, \quad 52 + 5 + 14.$$

Every total exceeds $e(Q) = 56$.

Now suppose that $\delta(G) = 3$. The same argument gives

$$e(G[U]) = 59, \quad A = 3, \quad Y = 3, \quad X = 0,$$

so $G = K_4 \dot{\cup} H$, where H has 22 vertices, 59 edges, and minimum degree at least three. Its minimum degree is at most five. If it is three, the eighteen nonneighbors induce at most $59 - 3 - 3 = 53$ edges; if it is four, the seventeen nonneighbors induce at most $59 - 4 - 6 = 49$ edges. Both possibilities contradict Lemma 4.1. If it is five, the sixteen nonneighbors induce at most $59 - 5 - 14 = 40$ edges, another contradiction to that lemma. $\square$

It follows that every color graph has minimum degree four. Choosing a degree-four vertex and applying Lemma 6.1 once more forces equality in (11):

$$G = K_5 \dot{\cup} H, \tag{12}$$

where H has twenty-one vertices, 55 edges, minimum degree at least four, independence number at most four, and is admissible. Its average degree is less than six, so $\delta(H)$ is four or five.

Excluding minimum degree four in the residual graph. Suppose that $\delta(H) = 4$. At a degree-four vertex, Lemma 4.1 forces equality in the neighborhood decomposition. Thus

$$H = K_5 \dot{\cup} Q,$$

where Q has sixteen vertices, 45 edges, minimum degree at least four, and independence number at most three. If $\delta(Q) = 4$, the eleven nonneighbors of a degree-four vertex induce at most $45 - 4 - 6 = 35$ edges and have independence number at most two, contrary to Lemma 3.2. Hence $\delta(Q) = 5$.

Choose a degree-five vertex v of Q , let $N = N_Q(v)$, let U be its ten nonneighbors, put $R = Q[U]$, and put $L = \overline{R}$. Then L is triangle-free and

$$e(R) \leq 45 - 5 - 14 = 26, \quad e(L) \geq 19. \quad (13)$$

If L is bipartite, Lemma 3.3(i) says that R consists of two K_5 's with matching cross-edges. Each vertex of N has at most one neighbor in either clique. Two nonadjacent vertices of N would then give an independent four-set in Q , as in the proof of Lemma 4.1. Thus $N = K_5$, making a forbidden K_6 with v . Therefore L is nonbipartite, and Lemma 3.3(ii) leaves only $e(L) = 19$ or 20.

If $e(L) = 19$, then $e(R) = 26$, and all counts are equalities:

$$A = 14, \quad Y = 6, \quad X = 8.$$

Let $M = \overline{Q[N]}$. It has four edges, and each vertex $a \in N$ has exactly $d_M(a)$ neighbors in U . By Lemma 3.3(iv), $\alpha(L) \leq 4$, so $\tau(L) \geq 6$. For any edge $aa' \in E(M)$, the two U -neighborhoods must cover L to prevent an independent four-set in Q , but their total size is at most $d_M(a) + d_M(a') \leq 5$. This is impossible.

It remains that $e(L) = 20$. Then $e(R) = 25$, and Lemma 3.3(iii) says that $L = T$, the balanced two-fold blow-up of C_5 . Here $A = 15$ and $Y \in \{5, 6\}$.

If $Y = 5$, then $M = \overline{Q[N]}$ has five edges, $X = 10$, and the cross-degree x_a of each $a \in N$ equals $d_M(a)$. Some edge aa' of M satisfies

$$d_M(a) + d_M(a') \leq 5. \quad (14)$$

Indeed, otherwise every edge would have endpoint degree sum six and hence would meet all four other edges. A pairwise-intersecting family of edges in a simple graph is contained in a star or a triangle, neither of which has five edges on five vertices. The two U -neighborhoods of the edge in (14) cannot cover T , contradicting $\tau(T) = 6$.

If $Y = 6$, then M has four edges and $X = 9$. Minimum degree gives $x_a \geq d_M(a)$ for each a , with total excess one; let z carry the excess. Every edge of M must meet z , since the endpoints of any other edge have at most five U -neighbors and cannot cover T . Hence $M = K_{1,4}$, centered at z , with $x_z = 5$ and cross-degree one at each leaf. For each center-leaf edge, the two neighborhoods form a minimum six-vertex cover of T and are disjoint. These four minimum covers all contain the fixed five-set $N_Q(z) \cap U$. Distinct minimum covers of T intersect in at most four vertices, so the covers coincide. The four leaf neighborhoods are therefore the same singleton, say $\{u\}$. The six-set consisting of v , u , and the four leaves spans fourteen Q -edges: four from v , six among the leaves, and four from u . This violates admissibility. We conclude that $\delta(H) \neq 4$.

Excluding minimum degree five in the residual graph. We are left with $\delta(H) = 5$. Choose a degree-five vertex v , let $N = N_H(v)$, let U be its fifteen nonneighbors, put $R = H[U]$, and put $L = \overline{R}$. Then

$$e(R) \leq 55 - 5 - 14 = 36. \quad (15)$$

If L is three-partite, each of its parts has size five. The graph R then consists of three K_5 's with at most six cross-edges. Every vertex of N has at most one neighbor in each clique. If two vertices of N were nonadjacent, each clique would supply three candidates avoiding both. At most six cross-edges rule out at most eighteen of the twenty-seven candidate triples, leaving an independent triple. Together with the two vertices of N , it gives an independent five-set in H . Thus $N = K_5$, which makes a forbidden K_6 with v .

Consequently L is not three-partite. Theorem 2.2 gives $e(R) \geq 34$, and Lemma 2.3(ii) excludes equality. Hence $e(R)$ is 35 or 36.

Suppose first that $e(R) = 35$. Then $A = 15$ and $Y \in \{5, 6\}$. Lemma 5.1 gives

$$R = C \dot{\cup} R_0,$$

where $C = K_5$ and $T = \overline{R_0}$ is the balanced two-fold blow-up of C_5 . Every vertex of N has at most one neighbor in C .

If $Y = 5$, then $M = \overline{H[N]}$ has five edges, $X = 10$, and every cross-degree equals the corresponding degree in M . Choose an edge $aa' \in E(M)$ satisfying (14), and choose $c \in C$ avoiding both endpoints. To prevent a, a', c together with an uncovered edge of T from forming an independent five-set, the two R_0 -neighborhoods must cover T . Their total size is at most five, contrary to $\tau(T) = 6$.

If $Y = 6$, then M has four edges and the cross-degrees have one unit of excess over the degrees in M . The same cover argument forces M to be $K_{1,4}$ and puts the excess at its center. For each center-leaf edge, all six cross-neighbors must lie in R_0 , since they are needed to cover T . The minimum-cover intersection property again makes the four leaf neighborhoods a common singleton. The vertex v , that singleton, and the four leaves span fourteen edges, a contradiction.

Finally suppose that $e(R) = 36$. All counts are equalities:

$$A = 14, \quad Y = 6, \quad X = 8.$$

The graph $M = \overline{H[N]}$ has four edges, and each cross-degree is the corresponding degree in M . Choose any edge $aa' \in E(M)$; its endpoint degree sum is at most five.

Apply Lemma 5.2. In alternative (i), let C be the K_5 , let c_0 be the endpoint in C of its unique edge to R_0 , and let $T = \overline{R_0}$. Each of a, a' has at most one neighbor in C , so there is a vertex

$$c \in C \setminus (\{c_0\} \cup N_H(a) \cup N_H(a')).$$

The vertex c has no R -neighbor in R_0 . The two R_0 -neighborhoods of a, a' must therefore cover T , but their total size is at most five and $\tau(T) = 6$.

In alternative (ii), choose c in the isolated K_5 avoiding a, a' . Now $T = \overline{R_0}$ has independence number at most four, so $\tau(T) \geq 6$. The same uncovered-edge argument again requires a vertex cover of size at most five. This final

contradiction proves that the graph H in (12) cannot exist, and completes the proof of Theorem 1.1.

8. CONCLUDING REMARKS

The proof is confined to five colors. It does not establish the general Erdős–Gyárfás conjecture, and the higher-color cases are not addressed here. The argument above is entirely graph-theoretic. In particular, no finite-search certificate or assumption about a structured coloring family enters the proof.

APPENDIX A. DEPENDENCY AUDIT

The external inputs are Brooks’s theorem and the Kang–Pikhurko extremal theorem, including its equality description for $(r, n) = (2, 10)$ and $(3, 15)$. All other finite structure statements used above are proved in the text.

Minimum degree is inherited only after equality in Lemma 2.4 gives $X = 0$ and peels off an isolated clique. Every later minimum-degree argument chooses a new minimum-degree vertex in the graph under consideration. At order fifteen, the possibility that the complement is three-partite is treated separately; at order ten, the possibility that the triangle-free complement is bipartite is also treated separately.

The main arithmetic identities are

$$5 \cdot 65 = 325 = \binom{26}{2}, \quad A \geq 14 \text{ when } d = 5,$$

and the two near-extremal splits are

$$35 - 4 = 31, \quad 36 - 4 = 32.$$

For reference, the proof dependencies are as follows.

- D1. Corollary 1.2 uses Theorem 1.1, the color-complement equivalence, and the affine-plane construction on $\mathbb{F}_5^2$ for the lower bound.
- D2. Proposition 2.5 uses the local bounds (1), Brooks’s theorem, and Theorem 2.2 with four parts.
- D3. The terminal results in Section 3 use only the local six-set bound, elementary triangle-free arguments, and the two-part case of Theorem 2.2 with Lemma 2.3(i).
- D4. Lemmas 4.1 and 4.2 use the preceding terminal lemmas, Lemma 2.4, and the three-part case of Theorem 2.2.
- D5. Lemmas 5.1 and 5.2 use Lemma 3.3 and recheck the bipartite complement branch before applying its nonbipartite alternatives.
- D6. Lemma 6.1 uses the independence-three obstructions and, at its terminal fifteen-vertex stage, Lemma 5.1.
- D7. Proposition 7.1 combines Proposition 2.5, Lemma 2.4, and Lemma 6.1; the identity $5 \cdot 65 = \binom{26}{2}$ then equalizes all color classes.
- D8. The final reduction uses the equality cases in the preceding lemmas, Lemmas 5.1 and 5.2, and the minimum-cover description in Lemma 3.3.

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